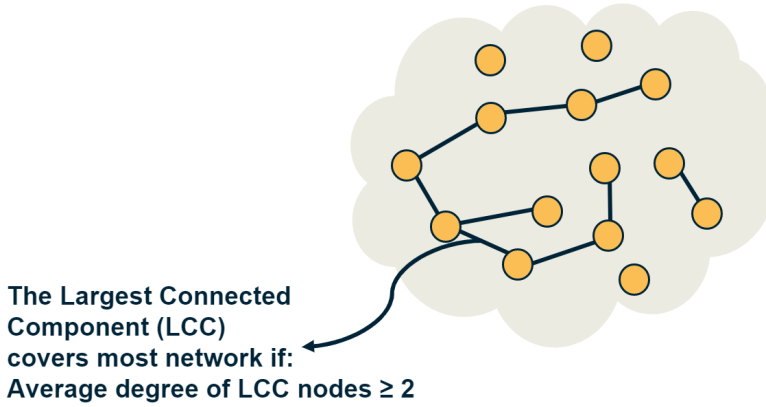


L11: Molloy-Reed Criterion



If the **largest connected component (LCC)** of a network covers almost all nodes, we refer to that LCC as the ***giant component*** of the network.

To derive the critical threshold of an inverse percolation process, we need a mathematical condition for the emergence of the giant component in a random network (*including power-law networks*). This condition is referred to as ***Molloy-Reed criterion (or principle)***:

Consider a random network that follows the *configuration model* with degree distribution $p(k)$

(i.e., there are no degree correlations). **The Molloy-Reed criterion states that the average degree of a node in the giant component should be at least two.**

Intuitively, if the degree of a node in the LCC is less than 2, that node is part of the LCC but it does not help to **expand** that component by connecting to other nodes. So, in order for the LCC to expand to almost all network nodes (i.e., in order for the LCC to be the giant component), the average degree of a node in the LCC should be at least two.

To express the Molloy-Reed criterion mathematically, let us first derive the average degree of a node in the LCC, as follows:

Suppose that j is a node that connects to the LCC through an LCC node i .

Let $P[k_j = k | A_{i,j} = 1]$ be the probability that node j has degree k , given that nodes i and j are connected.

From Bayes's theorem, we have that:

$$P[k_j = k | A_{i,j} = 1] = \frac{P[k_j = k, A_{i,j} = 1]}{P[A_{i,j} = 1]} = \frac{P[k_j = k] P[A_{i,j} = 1 | k_j = k]}{P[A_{i,j} = 1]} \quad (1)$$

For a network with n nodes and m edges, the denominator is simply:

$$P[A_{i,j} = 1] = \frac{2m}{n(n-1)} = \frac{\bar{k}}{n-1},$$

where $\bar{k}$ is the average node degree.

Also, $P[A_{i,j} = 1 | k_j = k] = \frac{k}{n-1}$ because j chooses between $n-1$ nodes to connect to with k edges (recall that the configuration model allows for multiple edges between nodes).

Returning to Equation-(1), we can now derive the average degree of nodes in the LCC:

$$\sum_k k P[k_j = k | A_{i,j} = 1] = \sum_k k \frac{P[k_j = k] P[A_{i,j} = 1 | k_j = k]}{P[A_{i,j} = 1]} = \frac{\sum_k k^2 p(k)}{\bar{k}} = \frac{\bar{k}^2}{\bar{k}}$$

So, returning to the Molloy-Reed criterion, we can now state it mathematically, as follows:

In a random network that follows the configuration model with degree distribution $p(k)$, if the first and second moments of the degree distribution are $\bar{k}$ and $\bar{k}^2$ respectively, the network has a giant connected component if

$$\frac{\bar{k}^2}{\bar{k}} \geq 2$$

Food For Thought

How does the Molloy-Reed criterion relate to the following result in Lesson-3 about $G(n, p)$ networks: a giant connected component emerges if the average node degree $\bar{k}$ is more than one? And how does it relate to the *average neighboring node*

degree $\bar{k}_{nn}$, we derived in Lesson-2?